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Review Article

Barriers and Facilitators to Engagement With a Weight Management Intervention in Asian Patients With Overweight or Obesity: A Systematic Review

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ABSTRACT

Objective: The obesity epidemic is a global health concern with Asian countries facing one of the most rapid rises in obesity rates. However, given the underwhelming long-term efficacy of weight loss strategies, especially in Asia, this review aimed to explore barriers and facilitators to weight management of patients with overweight and obesity in Asia.
Methods: Medline, CINAHL, PsycINFO, and Web of Science were searched for articles discussing barriers and facilitators of treatment to obesity from the perspectives of both health care professionals (HCPs) and patients. Qualitative and mixed method studies from Asia were included. Key quotes were extracted, coded, and thematically analyzed according to the methodology of Thomas and Harden.
Results: A total of 26 articles were included in this review. From patient perspectives, 3 main themes were identified: factors influencing poor eating behavior, inhibiting lifestyle modifications, and facilitating lifestyle modifications. Patients highlighted several barriers including the lack of social support, physiologic limitations to exercise, and low health literacy. Rigid sociocultural norms and lack of accessible health care services, exercise facilities, and healthy food exacerbated the barriers. Facilitators to lifestyle modifications consisted of strong support systems and high health literacy. HCPs agreed that low health literacy, lack of social support, and patient motivation impeded patients' weight loss attempts but were unaware of the other barriers they faced.

Abbreviations: BMI, body mass index; CASP, Critical Appraisal Skills Programme; HCP, health care professional.

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Conclusion: There are discrepancies between ideas of barriers and facilitators of HCPs and patients. A mixture of population level, primary care, and personal interventions are required to address this disparity, and enhanced health literacy can improve weight loss outcomes.

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Introduction

Global prevalence of obesity has tripled over the past 40 years, with an estimated one-third of the world's population being classified as those with overweight or obesity.[1,2] Asian regions, such as Southeast Asia, are expected to have an estimated 10%-20% rise in the prevalence of individuals with obesity.[3] This raises concern due to the burden of comorbidities associated with obesity, including the increased risk for cancer, cardiovascular diseases, type 2 diabetes mellitus, negative psychologic effects, and lower quality of life.[2,4,5]

The epidemic of obesity in Asia can be attributed to the "obesogenic" environment, created as a by-product of pursuing rapid economic growth, which facilitates overnutrition and decreased levels of physical activity.[6] There were numerous attempts at reducing the extent of obesity through weight loss and lifestyle modification programs.[7-11] However, studies have demonstrated that weight loss interventions, although effective in the short-term, lacked sustainability in the long run.[11,12] Up to 95% of individuals with obesity who managed to successfully lose weight regained their baseline weight within several years.[13]

In addition to the obesogenic environment, several other complex factors that are uniquely problematic for Asians have predisposed them to the development of adiposity-based chronic diseases. Compared with Westerners, Asians tend to have increased cardiometabolic risk at a lower body mass index (BMI), higher insulin resistance that increases the incidence of chronic diseases such as type 2 diabetes mellitus, and genetic predisposition to metabolic diseases at a younger age in certain Asian population groups.[14,15] Underlying biological mechanisms could induce an increase in food intake secondary to increasing energy expenditure during weight management interventions, counteracting the intended outcome.[16] Other biological factors including sex and age can determine an individual's response to weight loss management.[17] However, apart from a few rare genetic disorders associated with extreme obesity, research in genetic testing to offer personalized diet or activity plans is still in the early stages.[18] Conversely, sociocultural beliefs and personal factors play an important role too.[19] Hence, the objective of this review was to analyze the malleable societal, lifestyle, and psychologic barriers and facilitators to weight management.

To better comprehend the poor long-term efficacy of weight management interventions, it is necessary to understand the viewpoints of the recipients and health care professionals(HCPs) involved, with the goal of developing effective and sustainable weight loss policies.[9] Although previous studies have explored patient perspectives on the factors of success in weight management,[7] the perspectives of both HCPs and patients in the Asian context remains unknown. Thus, this study aimed to explore the barriers and facilitators of engagement to weight loss management in Asian patients with overweight or obesity.

Materials and Methods

Search Strategy

This article was written in accordance with the Preferred Reporting Items for Systemic Reviews and Meta-Analyses guidelines[20] (Fig. 1 and Supplementary Material 4). Medline, CINAHL, PsycINFO, and Web of Science were searched for articles discussing barriers and facilitators of engagement of weight management in March 2022. Key search terms such as "obesity," "overweight," "barriers," "facilitators," "compliance," and "adherence" and a qualitative filter were used. Articles were retrieved and duplicates removed using EndNote 20. The full search strategy is attached in Supplementary Material 1.

Eligibility and Selection Criteria

Four authors (V.V.A., E.L.C.Z, Y.H.C., C.L.) independently screened all titles and abstracts according to the eligibility criteria, which included the following: (1) articles discussing the barriers and/or facilitators of engagement of weight management of patients with overweight or obesity, (2) articles using qualitative methodology, and (3) peer-reviewed articles written or translated into the English language. Articles not describing the Asian perspective of weight loss or the weight management strategy were excluded. This review followed the World Health Organization (WHO) Asia classification of weight whereby a BMI of 23-27.4 kg/m^2 is considered overweight and a BMI $\geq$ 27.5 kg/m^2 is considered obesity.[21] Subsequently, 2 authors (V.V.A., E.L.C.Z.) independently screened the remaining included full-text articles and their respective supplementary materials to determine whether they met the inclusion criteria. Any discrepancies were discussed in consultation with a third independent senior author.

Data Extraction and Analysis

Data were independently extracted by 2 authors. Details of the articles (country, design methodology, and type of participants), participant demographics (sex, age, and BMI), and the study setting were included. Next, a thematic synthesis of key quotes was performed using the Thomas and Harden methodology for a thematic analysis.[22] Coding of the key quotes was performed thoroughly, transcripts were analyzed and compared, and similar quotes were categorized into preliminary descriptive themes by each reviewer independently. These initial themes were reviewed by all authors, modified, and refined. New codes were generated as per the newly defined themes, which were used in the eventual write-up. To ensure that the coding framework and themes captured all relevant data from the included studies, discussions were held among the authors in consultation with a senior author.

Quality Assessment

The articles were assessed using the Critical Appraisal Skills Programme (CASP) tool for a qualitative research by 2 authors independently, with discrepancies being resolved through consensus by a senior author.[23] The CASP checklist includes questions that broadly determine whether the results of the study are valid; if so, what the results of the study are, and whether they will help locally. Articles were not excluded based on the quality of the assessment; instead, the assessment allowed us to give appropriate weight to the findings of each study and improve the quality of data synthesis.[24]

Results

Summary of Included Studies

Our initial search strategy yielded 3843 articles, of which 26 were eventually included in the analysis (Fig. 1), comprising 1072 patients. The mean age ranged from 18 to 86 years. By region, the articles were from East Asia (n = 6), Middle East (n = 10), South Asia (n = 2), and Southeast Asia (n = 8).[25-50] The included studies used focus-group discussions (n = 11), semistructured interviews (n = 17), and an unstructured interview approach (n = 1). Studies examined the perspective of patients (n = 21),[25-32,34-37,39,41,43-48,50] medical professionals, comprising primary care physicians, pharmacists, nurses, dietitians, and doctors from obesity-related specialties (n = 4),[38,40,42,49] or both (n = 1).[33] Details of the included articles and quality assessment are appended in Supplementary Materials 2 and 3, respectively. Three main themes were identified from thematic analysis namely, patient factors influencing poor eating behavior, inhibiting lifestyle modifications, and facilitating lifestyle modifications. A summary of the themes and its corresponding quotes can be found in Table 1 and a graphical illustration

Highlights

- Main themes were poor eating behaviour and factors impacting lifestyle modifications.
- Barriers were functional limitations, lack of social support and low health literacy.
- Facilitators to weight management were strong support systems and health literacy.
- Health care professionals were unaware of emotional, functional and economic barriers.
- Discrepancies between ideas of health care professionals and patients need addressing.

Clinical Relevance

This study highlights challenges in achieving weight loss for individuals with obesity among Asians. There is an urgent need of educating people about obesity, to remove societal stigmatization, and mobilizing societal support for individuals in weight loss. An important step is to empower health care professionals with skillsets in managing obesity.

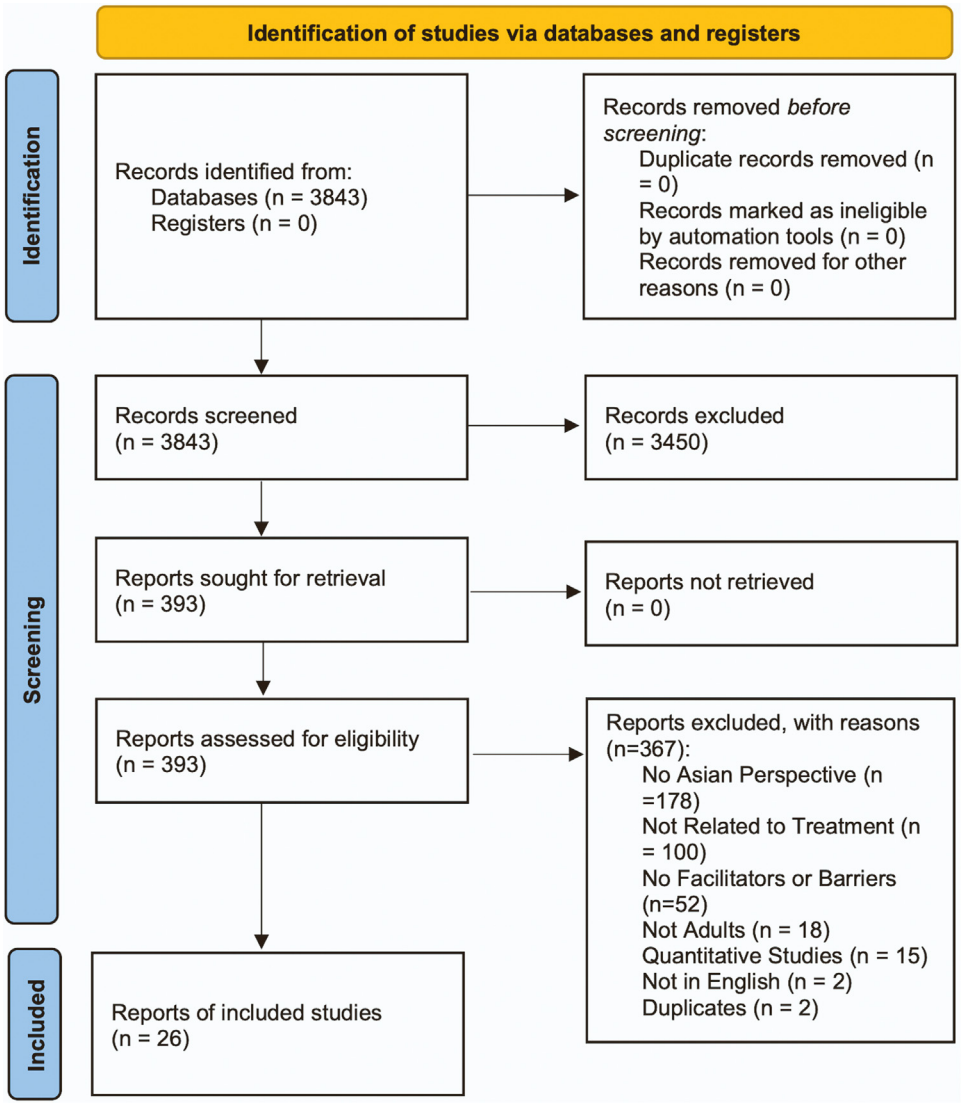

Fig. 1. Preferred Reporting Items for Systematic Reviews and Meta-analyses flowchart.

Table
Summary of Themes and Subthemes of the Barriers and Facilitators to Weight Loss

Headers	Themes with regional breakdown	Subthemes with quotations
Barriers influencing poor eating behavior	Sociocultural norms: 11 studies (SEA = 4, ME = 5, EA = 2, SA = 1)	Peer pressure "If we do not eat the food prepared by her [my wife], she will feel hurt, so I am forced to eat." "My coworkers bring coffee and sweets every single day and sharing [these treats with them] is one way to show my politeness" Societal beliefs "Thinness is a sign of weakness and disease, and it is an indication of having problems and being unhappy with spouse and family." "We love obesity, and no one is ridiculed for being obese."
	Accessibility to healthy food sources: 8 studies (SEA = 2, ME = 5, EA = 1)	Scarcity of healthy food "The restaurants near my workplace offer limited healthy choices." "First of all, you cannot easily find stores offering healthy foods." Ubiquity of unhealthy options "Unhealthy food is around me everywhere"
	Economic: 6 studies (SEA = 3, ME = 2, EA = 1)	High cost of healthy options "The problem with my diet was that it wasn't economical" "One of the barriers was the expensive product." Affordability of unhealthy options "The price is quite affordable and the taste ok."
	Emotional eating: 6 studies (SEA = 1, ME = 4, EA = 1)	Respite from stress "I'm stressed because my husband tells me to lose weight." "I relieved my stress by eating." "No time to eat lunch. Must finish housework because husband doesn't want to see a messy house. Stress, so we snack!"
Barriers inhibiting lifestyle modifications	Lack of personal motivation: 15 studies (SEA = 6, ME = 5, EA = 4)	Lack of energy "Sometimes I feel lazy and have no mood to do exercise. I just want to relax at home. I don't know why" "…actually if I want to do it, but the thing is, I feel so lazy" Lack of discipline "I am tired of keeping the diet. I do not like being disciplined also the regular visits" "I cannot control my desire to eat"
	Lack of support: 13 studies (SEA = 4, ME = 7, EA = 1, SA = 1)	Social support "The family might not be completely supportive in some occasions and complain about the diet." "My family would not encourage me to lose weight" Lack of company "I feel shy to jog alone in the park. It is boring doing it alone" "'I used to work out regularly with my friends. However, now they are unable to spend time at the gym. Therefore, I have no more company to work out at the gym. I cannot workout alone."
	Physical environment: 9 studies (SEA = 3, ME = 3, EA = 3)	Exercise facilities "Some of the facilities are old and broken, I am worried I will get hurt." "There are no parks or facilities to exercise near my house." Health care services "The health center was too far from my house." "It takes forever to get to see your doctor" Inclement weather "It is difficult to walk in hot weather" "Weather changes such as when it is hot or dusty often interrupts my weight loss program" Sedentary lifestyle "Most people are white-collar workers, and they don't need much physical power." "Many villagers shifted to the cities after the 1979 revolution, and urban facilities became better. Because urban life was easier, they didn't need much physical work."
	Low health literacy: 8 studies (SEA = 2, ME = 3, EA = 2, SA = 1)	Lack of information "The majority of people do not have a plan for obesity prevention, and if they do, they do not know what to do exactly." "I don't know much about diets; I consume whatever I know that it is right." Misconceptions "I did not expect obesity, while I was an overeater. All I wanted to do was to satisfy my hunger regardless of nutritional value of what I was eating. In fact my diet was not healthy" "After losing weight, I thought I no longer needed to be on a diet"
	Physiologic limitations: 8 studies (SEA = 5, ME = 3)	Adverse effects "I feel depressed when I start dieting" "I just consumed it for 1 month. I could not stand it. The drink made me nauseated that I feel like vomiting." Preexisting conditions "I want to exercise but I can't. I can't jog. My knees are painful." "When I was young, I would exercise at least twice a week. However now, I have to stop after the problem with my knee."

(*continued on next page*)

Table (*continued*)

Headers	Themes with regional breakdown	Subthemes with quotations
Facilitators of lifestyle modifications	Support systems: 11 studies (SEA = 4, ME = 3, EA = 4)	Social support "It is important to get encouragement from family members." "My motivation comes from my family, especially my wife." Role models "I want to be a role model for my family." "I want to also set a good example for them (children)." Expertise of health care professionals "My experienced coach would give me the right instruction and positive feedbacks, which were very helpful and kept me motivated to continue exercising with more energy and vitality." "An expert's support and motivation can be very effective on outcome"
	Health literacy: 10 studies (SEA = 2, ME = 3, EA = 4, SA = 1)	Knowledge of healthy options "I am confident I can lose my weight if I know the right way to do it." "Knowledge of different foods and their calories could be the most important motivator" Knowledge of complications of obesity "So it just made me think that before I hit a point of no return, I better do something about it." "We can avoid a lot of diseases like heart disease and high cholesterol level."
	Personal motivation: 8 studies (SEA = 3, ME = 4, EA = 1)	Self-discipline "If you want to be healthy you must force." "I came to the conclusion that I need to help myself." Positive feedback loop "As I see the result, it encourages me to exercise more and more, until I reach my desired weight." "When I saw the positive physical and mental outcomes of losing weight, I felt a strong motivation to keep my regimen." Desire to improve appearance "Because I want to attract someone." "Having thin body will give me more options in clothing."

SEA, South East Asia; ME, Middle East; EA, East Asia; SA, South Asia.

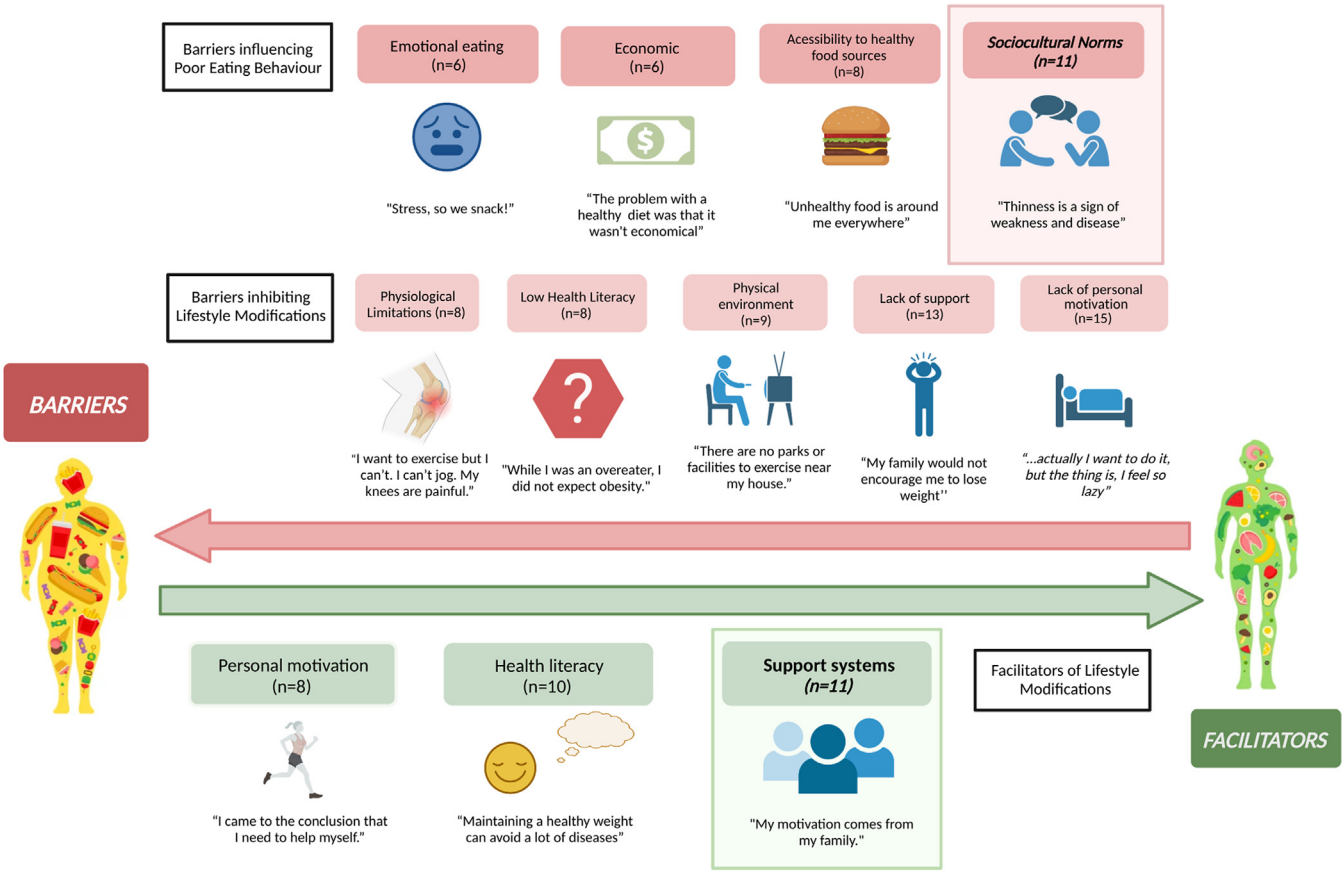

Fig. 2. Graphical representation of themes and subthemes of barriers and facilitators to weight loss. n, number of articles.

of the themes in Figure 2. The number of studies supporting the theme and a regional breakdown of the articles are indicated.

Patient Factors Influencing Poor Eating Behavior

Sociocultural Norms

Patients trying to lose weight had to deal with deeply entrenched sociocultural norms such as peer pressure to eat unhealthily,[25,32,36,37,39,44,46,50] cultural norms toward diets, and being obese.[33,36,39,41,46,47] Patients found it hard to resist unhealthy food served during special occasions.[25,36,39,44,46,50] Moreover, they did not want to offend family and friends by not eating offered food.[25,36,39,44,46,50] Patients were brought up in societies with indoctrinated views that rice, bread, and desserts were main staples of every meal[33,36,41,46,47] or that food should be consumed in large quantities and not be wasted.[33,39,47] Furthermore, a considerable number of patients lived in societies where obesity was promoted as a sign of status, whereas thinness was synonymous with weakness.[33,35]

Accessibility to Healthy Food Sources

The lack of healthy food options,[25,31,32,36,47] the ubiquity of unhealthy food,[25,37,44,47] and the absence of appropriate food storage facilities were major barriers to healthy dietary lifestyles.[35] Many restaurants did not offer many healthy food choices,[25,32,36,47] and there was inconvenience in finding and preparing healthy food.[25,31,32,36] Most patients felt that unhealthy food choices were more accessible, making it even harder for them to maintain a healthy diet.[25,37,46,47] In addition, workplaces often lacked the equipment to store premade meals.[35]

Economic Factors

Patients generally agreed that losing weight could be expensive,[29,32,35,44,50] and it would be cheaper to eat unhealthily.[46] A study in Malaysia highlighted one of the crucial themes underlying availability and accessibility to healthier foods is the affordability of outside food that were cheaper, were more delicious with more variety, but were unhealthier and high in calories, fat, and salt.[46] The exorbitant cost of approved weight loss products, fitness equipment, and healthy food made it hard for them to prioritize weight management when they were already struggling with other financial responsibilities.[29,32,35,44,50]

Emotional Eating

A significant number of patients expressed emotions such as stress, distress, anxiety, and frustration in their daily lives.[29,32,36,37,39,50] These people often turned to snacking "comfort food" as a form of seeking respite from emotional pain.[39,50] A few who were on dietary modification and other weight management programs found themselves curtailing treatment because of the emotional suffering.[32,36,37]

Patient Factors Inhibiting Lifestyle Modifications

Personal Motivation

A significant proportion of patients saw their lack of motivation and energy or self-discipline as a significant barrier to weight loss.[25,26,32,35,36,39,43,46,47,50] They believed that their motivation to commit to a dietary change or regular exercise was unsustainable and often faltered once they were tired or faced with inconveniences.[26,28,29,31,32,43,44] Many patients felt it was challenging to constantly resist cravings for snacks, resulting in relapse into unhealthy eating habits and discontinued exercise regimes.[32,35,36,39,43,44,46,47,50] Moreover, they commonly perceived work and academic commitments and familial responsibilities to be of greater priority than losing weight.[25-27,29,32,35,39,43-46,50]

Lack of Support

Patients undergoing weight management programs were usually plagued by a lack of support from their friends and/or family,[29,32,33,35-37,39,41,44-47] institution, or workplace.[31,32] Difficulty in finding a companion to exercise with[29,39,44-46] and unsupportive family and friends[29,35-37,39,43,47,50] contributed to their hesitancy toward exercising. Some even received demoralizing comments while attempting to lose weight, resulting in feelings of frustration.[32,33,35-37,39,47] A few felt neglected by their employers' lack of focus on their health and physical wellbeing.[31]

Physical Environment

The lack of access to exercise facilities[29,33,36,46] and health care services,[28,29] together with unpredictable weather[26,36] and the transition toward a sedentary lifestyle[31,33,47,50] were significant barriers to weight loss. Patients felt discouraged to visit health care facilities for weight loss because of the travel distance and long waiting times.[28,29] The introduction of technology that simplify and automate many aspects of daily living, such as automobiles for traveling, may have promoted a sedentary lifestyle with reductions in physical activity and, hence, potential weight gain.[31,33,47,50]

Low Health Literacy

A common theme surrounding barriers to weight loss was the lack of information[29,33,43] and the misconceptions patients had.[27,32,33,35,36,41,46] Some patients were unaware of the exercise and health care facilities that were available to them[29,33] and had little knowledge on proper exercise and diet.[29,33,43] Notably, some misperceived their current lifestyle to be healthy because they were following nonprofessional advice (ie, skipping meals) or believed that their weights were at normal levels.[27,32,33,35,36,41,46] Patients who exhibited the view that weight loss was a one-off effort rather than a long arduous journey often failed to sustain the weight loss programs and goals.[25,26,32,33,43,44,46,47,50]

Physiologic Limitations

The adverse effects associated with many patients' weight loss plans obstructed their progress in reducing their weight.[31,32,36,43-46,50] Some patients felt a deterioration in their physical state, claiming that the new dietary plans resulted in a general sense of weakness and, hence, deterred them from completing their dietary modification.[36,43,44] Other adverse reactions related to their new dietary modification included constipation, nausea, and gastric pain.[32,43-45] Some shared their challenges in undergoing certain exercise regimes because of preexisting medical conditions and/or previous injuries that limited their ability to exercise, including knee injuries and osteoarthritis.[31,32,44-46,50]

Facilitators of Lifestyle Modifications

Support Systems

Many patients attributed their success in weight loss to existing social support systems[26,30,34-36,44-47] and good role models.[44,45] They often found encouragement from people close to them having substantial positive effects on their motivation to losing weight.[35,36,45,46] Patients revealed that peer support groups were useful in overcoming their negative thoughts and doubts.[26,34,35,37] Several patients credited their success to the advice and expertise of HCPs.[26-28,46] HCPs who adopted an encouraging attitude significantly improved patients' motivation to continue their weight loss regime.[25,26,34,35] Moreover, some patients were influenced by

extrinsic feedback, such as improving their physical appearances,[26,36,44-46] and being a role model[26,36,47] to others.

Health Literacy

The knowledge on healthy options for weight loss[26,27,35-37,41,46] and the fear of complications associated with obesity was a significant motivator for losing weight.[25,26,28,36,41,46,47] A thorough understanding of the ideal strategies for weight loss empowered patients to lose weight more efficiently.[26,27,36,46] Next, understanding the physical implications of obesity,[25,28,36,46,47] from a range of chronic conditions from diabetes[28,46] to joint and mobility issues,[25,36,46] can evoke concern toward potentially developing these complications. This can be redirected as a form of motivation for weight loss.[25,26,28,36,41,46,47]

Personal Motivation

Personal motivation was identified as an important factor in achieving patients' weight loss targets. Many believed that self-discipline was the key.[28,34,35,45-48] Furthermore, patients felt motivated from seeing substantial progress in weight loss,[31,34-36,45] and this was associated with improvements in mental and spiritual wellbeing, which in turn helped with maintaining their weight loss progress.[31,35]

Provider Perception on Weight Loss

Provider Perception of Patient Barriers and Facilitators of Weight Loss

HCPs identified low health literacy,[38,40] personal motivation,[38,40,49] and lack of familial support[40] as major barriers to patients successfully persisting with weight loss measures. The preference of a sedentary lifestyle,[40,42,49] along with a lack of accessible exercise facilities[40] and weight management programs,[40,42] exacerbated the situation. Moreover, sociocultural norms,[40] language barriers,[38] and general unwillingness to comply with professional advice[38] made patients reluctant to adopt healthier behaviors.

Conversely, HCPs perceived patients with good health literacy, such as knowledge of healthy options[40,42] and the adverse effects of obesity,[49] demonstrated better compliance with weight loss advice. Patients who were more conscious of their body image[40] and had a strong support system[38,40,49] had far more determination to continue losing weight than those without them.

Issues Associated With Providing Weight Loss Services

Many HCPs admitted they were understaffed[40] or were not provided with resources[38,40] (ie, brochures and dedicated clinics) to help patients lose weight. They claimed there was a lack of knowledge to provide professional dietary advice.[38,42,49] Furthermore, collaborative efforts with other health care providers such as pharmacists seemd insufficient, making it difficult for them to follow-up with patients' weight loss efforts.[38] However, when a functional multidisciplinary team (involving dietitians, social workers, physicians, nurses, and fitness trainers) was available, many found that it vastly improved patients' weight loss experience.[38,40,42]

Discussion

The overall burden of overweight and obesity is predicted to reach 57.8% of the world's population by 2030.[51] Despite the concerted global interventions, many patients who are overweight or obese either fail to lose weight or regain the weight shortly after their treatment.[13] This study analyzed 26 articles, with an average CASP score of 8.7 and only 2 papers with a score of 7, indicating a high level of reliability in the evidence synthesized. We found that patients failed to lose weight owing to numerous factors contributing to the failure of dietary and lifestyle modifications. Of interest, the synthesized themes are closely linked to the 5 components of the social determinants of health, namely economic stability; education, social, and community context; health and health care; and neighborhood and built environment.[52] Studies have shown that increasing burden of social determinants of health is tied to rising obesity levels.[53,54]

Weight loss strategies are often centered on an individual's understanding and actions toward curbing calorie surplus through dietary and lifestyle measures.[55,56] However, we have highlighted numerous barriers impeding the effectiveness and sustainability of these efforts. Asians had to grapple with deeply entrenched sociocultural norms that encourage consuming large quantities of unhealthy food, placing them in a predicament between adherence to their family's cultural norms and their individual weight loss efforts.[25,29,31-33,35-37,39,41,44-47,50] In addition, a divergence in societal beliefs between the Middle East and Asia was observed. Participants from the Middle East were more likely to report positive views toward obesity, and women living there faced expectations of being homemakers, denying them opportunity to leave their homes for exercise.[33,36,37,40] A postulated reason is the residual patriarchal ideology in the Middle East.[57] Therefore, addressing the cultural norms in these countries is paramount in implementing health care policies targeting lifestyle interventions. Furthermore, there was a lack of social support in patients' weight loss efforts. This is exacerbated by the socially oriented culture in the East, where values such as filial piety and the communal spirit are emphasized, when compared with the more individualistic Western culture.[58,59] Thus, interventions should be culturally sensitive to tap on the strength of this socially oriented culture to support these patients through their weight loss.

For most Asian countries, the topic of weight loss is taboo, and the stigma surrounding obesity may bring shame to the individual.[60] Addressing the weight stigma should be initiated as early as during childhood because this is a critical time for intervention given the significant association with obesity in the later years. Places such as schools are suitable environments for intervention, with the promotion of obesity preventive programs that discourage weight-teasing and weight discrimination.[61] Weight-stigma reducing interventions make up a vital component in the holistic reduction of childhood obesity, and several institutions have recommended the application of it.[62] It is part of a nationwide multipronged approach targeting obesity by altering children's lifestyles and behaviors because these changes persist throughout their lives.[12,63] These efforts have been observed in Singapore, China, and many other East Asian countries.[64] Schools play a crucial role by increasing physical activities and health education classes to address misconceptions surrounding obesity.[65,66] Incorporating social support from friends and family in weight loss management has been shown to improve results.[67] Hence, interventions involving social groups at its core could prove to be effective in family-centric Asian communities. Moreover, efforts in improving health literacy should be the focus for health promotion strategies. This study highlighted 9 studies reporting the patients' misperceptions of their lifestyle as healthy or weight as normal. This may have negative long-lasting implications on their perception on the barriers of facilitators to weight loss and their motivation to pursue positive lifestyle modifications.

Gradual urbanization and dietary transition of Asian countries have led to sedentary lifestyles and dependence on fast food for meals.[68] There is a gradual deviation from traditional diets, which reduce the risks of chronic metabolic disease and weight gain, negatively affecting weight loss and metabolic health.[69,70]

Suggestions of taxing unhealthful options have proven not to be effective in promoting weight loss. Rather, efforts in improving the affordability of healthful food choices and encouraging the use of traditional and local diets may be more helpful in altering patients' perception of cost disparities in food availability.[65,69,71,] In addition, the implementation of health policies encouraging physical exercise on the population level can be effective. Several countries have explored the use of physical challenges using phone applications to incentivize physical activities, such as LOSE IT to WIN IT challenge in the United States, the National Steps Challenge in Singapore, or the HeadUp app in the United Kingdom.[72-74] Although these activities brought only modest improvement in weight management, creative incentives have potential in addressing certain barriers to weight loss perceptions.[72-74]

The American College of Cardiology and the American Heart Association recommend that physicians assess a patient's readiness to make lifestyle changes and a patient's weight and health goals and to identify barriers to achieving weight loss before recommending individualized intervention strategies for patients.[75] The perception of HCPs is paramount because their direct involvement in the medical assessment, counseling, and coordination of lifestyle modifications is heavily associated with the patient's success.[76] Our study found that although HCPs identified most barriers and facilitators, they failed to identify the sociocultural issues, physiologic limitations, or specific personal motivations to weight loss. In addition, HCPs felt ill-equipped to discuss the sensitive topic of patients' weight. There were also conflicting views on whether weight management was under their purview.[77] At the systems level, this discrepancy in views between the HCPs and patients could impede their clinical judgments, leading to unequal access to care.[78] Similar to other Western studies,[79] HCPs have identified patients' shortcomings (ie, health literacy and motivation) as more significant[40,42,49] than the lack of social support.[40] However, it is important to note that the disparity between the perceptions of HCPs and patients might be interpreted as a form of bias on the part of the health care professional. Apart from addressing the patient factors, the establishment of a strong support system remains one of the priorities in promoting compliance to lifestyle modifications. Moreover, studies have uncovered that almost one-third of HCPs may have an innate bias toward patients with obesity, which can influence patients to avoid appointments and underreport concerns during consults, leading to poorer outcomes.[80,81] HCPs should be cognisant of their weight bias and adopt nonstigmatizing language to improve the outcomes of patients.[82] Other shortcomings include the lack of staff and resources for providing care for individuals with obesity, such as bigger and stronger beds to transport patients across different departments.[83] More resources should be channeled to address the rising obesity rates. Finally, a multidisciplinary approach to weight loss, including a dietitian, social worker, and pharmacist, should be considered in all weight loss programs.[84,85]

Strengths and Limitations

This study is the first to comprehensively summarize the unique perspectives of patients and HCPs on the barriers and facilitators of engagement of weight loss management in Asia. However, this study has its limitations. First, the results obtained from the semi-structured interviews were subject to the interpretation by the interviewees, and many studies did not include quantitative data such as mean BMI. Second, the articles reviewed did not address the barriers toward other weight loss interventions such as surgery or pharmacotherapy, which will be highly valuable for future studies. Third, no quantitative measurement of weight loss was provided, and the assumption that any evidence stating weight loss equated to successful weight loss needs to be made with caution. Fourth, there may have been biases in interview responses in the form of unconscious compliancy to societal norms. Furthermore, there is limited representation of male perspectives given a sizable number of included studies are women-only studies (n = 8). Fifth, the current findings may have been skewed toward the Southeast Asia (n = 8) and Middle East (n = 10), and they may not be generalizable beyond the subpopulations that the studies were conducted in. Moreover, differences exist between various Asian countries regarding the phenotypical traits and different obesity or metabolic disease manifestations. However, this study can serve as the first step into recognizing the various issues that patients with overweight or obesity may face in their weight management journey in Asia. Future studies are warranted to examine the interethnic and cultural differences in the perspective and behaviors toward interventions for weight loss within the Asian population.

Conclusions

This study highlights the challenges in achieving weight loss for individuals with obesity among Asians. There is an urgent need of creating awareness to educate people about obesity, to remove societal stigmatization, and to mobilize society to support individuals in weight loss. An important step is to empower HCPs with skillsets in managing obesity. A safe facility consisting of a multidisciplinary health care team will be ideal to support patients with obesity.

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Author Contributions

V.V.A., E.L.C.Z, and Y.H.C. are co-first authors. All authors have made substantial contributions to the following: (1) the conception and design of the study, acquisition of data, or analysis and interpretation of data, (2) drafting the article or revising it critically for important intellectual content, and (3) final approval of the version to be submitted. No writing assistance was obtained in the preparation of the manuscript. The manuscript, including related data, figures and tables has not been previously published and that the manuscript is not under consideration elsewhere.

Data Availability

All articles cited in this manuscript are available from Medline, CINAHL, PsycINFO, and Web of Science.

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